

Aliah University

Natural Language Processing (MCAPGPE13)

MCA 4th Sem

Spring 2026

Full Marks -80 Time- 3:00 Hrs

Group A

Answer all questions [10*1=10]

1. Select the best alternative

- i) Which of the following is common NLP application ? <CO1 , L1>
(A) Machine translation (B) Sentiment analysis
(C) Speech recognition (D) All of the above
- ii) Word embeddings are used to represent words as: <CO2 , L1>
(A) Character array (B) Numerical vectors
(C) Linked list (D) Binary trees
- iii) NER stands for: <CO5 , L1>
(A) Natural Entity Retrieval (B) Named Entity Recognition
(C) Neural Embedding Representation (D) Normalized Error Rate
- iv) In Hidden Markov Model for POS tagging, the hidden states usually represent: <CO3 , L1>
(A) Words (B) POS tags
(C) Documents (D) Tokens
- v) In Information Retrieval, an inverted index maps: <CO4 , L1>
(A) Documents to page ranks (B) Terms to documents containing them
(C) Documents to queries (D) Sentences to grammar rules
- vi) What is the full form of NLU? <CO1 , L1>
(A) Nature Language Understanding (B) Neutral Language Understanding
(C) Natural Language Understanding (D) Non-human Language Understanding
- vii) What is the study of organizing sound systematically? <CO1 , L1>
(A) Phonology (B) Morphology (C) Morpheme (D) None
- viii) Machine Translation converts - <CO1, L2>
(A) Human language to machine language (B) One human language to another
(C) Any human language to English (D) Machine language to human language
- ix) Which of the following is NOT a text preprocessing step <CO5 , L2>
(A) Stemming (B) Word-Embedding (C) Tokenization (D) Lemmatization
- x) Natural Language Processing (NLP) is a field of <CO1 , L1>
(A) Artificial Intelligence (B) Computer Science (C) Linguistics (D) All of the above

Group-B

Answer any 4 questions [4*5=20]

1. Differentiate between Stemming and Lemmatization with proper examples. <CO5 , L5>
2. Explain TF-IDF word embedding mentioning significance of its score. <CO3 , L4>
3. Compare between rule-based and stochastic taggers. <CO3 , L5>
4. Describe utility of NLP in different domains. <CO1, L2>
5. Differentiate between Boolean and Vector models in Information Retrieval. <CO5, L4>

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Group-C

Answer any 5 questions [5*10=50]

7. Explain the Single link and Single pass algorithms for document clustering with suitable examples. [5+5] <CO4, L2>
8. What is N-gram model? Explain Word2vec model with a suitable example. [4+6] <CO2, L4>
9. Find the POS of the following sentence using Viterbi Algorithm. [10] <CO3, L3>

The<DT> coaches<NN/VB> train<NN/VB> the<DT> team<NN/VB>

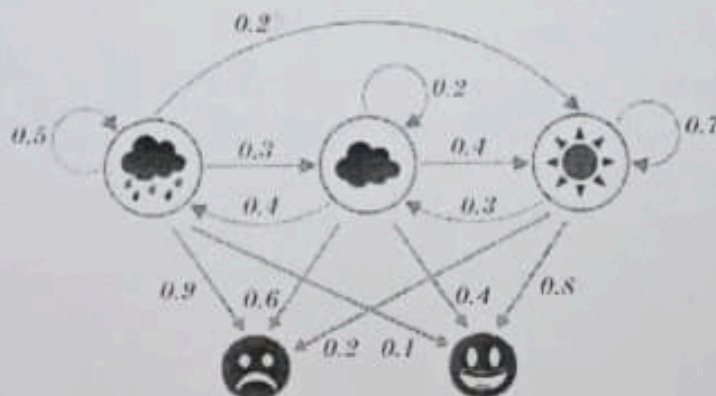
Emission Table

Word	DT	NN	VB
The	0.90	0.02	0.01
coaches	0.01	0.65	0.25
train	0.01	0.35	0.55
the	0.90	0.02	0.01
team	0.01	0.70	0.20

Transition Table

	DT	NN	VB
<start>	0.8	0.2	0
DT	0.05	0.85	0.10
NN	0.30	0.20	0.50
VB	0.65	0.25	0.10

10. What are the common challenges in processing natural languages? Explain different types of ambiguities with example. [2+8] <CO1, L3>
11. Consider the observed variables as Happy and Sad, and hidden variables as Rainy, Cloudy, and Sunny. Also consider the stationary probability of <rainy, cloudy, sunny>=[0.218,0.273, 0.509]. Find the probability P (sunny, cloudy, rainy| happy, sad, sad) using Hidden Markov Model. [10] <CO3, L3>



12. Derive Naive Bayes theorem from conditional probability. Calculate the probability P(No|Rainy) using the Naive Bayes theorem on the following data. [3+7] <CO3, L3>

Weather	Play
Sunny	No
Overcast	Yes
Rainy	Yes
Sunny	Yes
Sunny	Yes
Overcast	Yes
Rainy	No
Rainy	No
Sunny	Yes
Rainy	Yes
Sunny	No
Overcast	Yes
Overcast	Yes
Rainy	No